**Table S1:** A four-point example in a single cell. Model A moves every case to the cell's label frequencies; model B moves each case towards its own correct label by the same amount. The two models are equally far from $q_0$, and only B uses information about the individual case.

| case $i$ | label $y_i$ | $q_0(\cdot \mid x_i)$ | model A: $q_1(\cdot \mid x_i)$ | model B: $q_1(\cdot \mid x_i)$ |
|---|---|---|---|---|
| 1 | 1 | $(0.5, 0.5)$ | $(0.75, 0.25)$ | $(0.75, 0.25)$ |
| 2 | 1 | $(0.5, 0.5)$ | $(0.75, 0.25)$ | $(0.75, 0.25)$ |
| 3 | 1 | $(0.5, 0.5)$ | $(0.75, 0.25)$ | $(0.75, 0.25)$ |
| 4 | 2 | $(0.5, 0.5)$ | $(0.75, 0.25)$ | $(0.25, 0.75)$ |

# Supplementary material

## WAGER: An Exact Decomposition of Model-Pair Score Gains on Benchmarks with Strong Label Priors

This document contains the proofs, the influence-function derivation, the implementation and reproducibility details, and the secondary empirical studies referred to in the manuscript. Section, equation, table and figure numbers are prefixed with S throughout, and the manuscript cites them in that form. Numbered results stated in the manuscript are restated here before being proved, so this document can be read on its own.

### S1. A worked example

Four data points are enough to show the whole construction, and to exhibit in arithmetic the main properties Section 3 of the manuscript proves in general. Take one cell containing four test cases and two possible labels. Three of the cases carry label 1 and one carries label 2, so the cell's label frequencies are $(3/4, 1/4)$. The old model $q_0$ is uninformative within the cell: it predicts $(0.5, 0.5)$ for all four cases. We compare two candidate replacements, shown in Table S1.

Score both models with the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, and for each case tabulate the score *difference* at each of the two possible labels, not only at the observed one. Writing $h_i(y) = S(q_1(\cdot \mid x_i), y) - S(q_0(\cdot \mid x_i), y)$, all the arithmetic is in eighths:

$$h_i(y) = \begin{cases} +3/8 & \text{if } y \text{ is the label the new model moved towards,} \\ -5/8 & \text{otherwise.} \end{cases} \tag{S1}$$

This vector, evaluable at labels that were never observed for case $i$, is the only object the method needs beyond the labels themselves.

*Model A: a pure prior refit.*. Every case gets the same prediction, so $h_i(1) = +3/8$ and $h_i(2) = -5/8$ throughout, and the observed mean gain is $\Delta T = \frac{1}{4}[3 \cdot \frac{3}{8} - \frac{5}{8}] = \frac{1}{8}$. Now transport: replace each case's label in turn by each of the other three and average. Cases 1–3 are each scored against $\{1, 1, 2\}$, giving $\frac{1}{3}(\frac{3}{8} + \frac{3}{8} - \frac{5}{8}) = \frac{1}{24}$; case 4 is scored against $\{1, 1, 1\}$, giving $+\frac{3}{8}$. So $\Delta P = \frac{1}{4}[3 \cdot \frac{1}{24} + \frac{3}{8}] = \frac{1}{8}$ and $\Delta R = 0$ — exactly, in this sample, not on average over hypothetical resamples. That is the general fact: a prediction change constant within a cell contributes nothing to $\Delta R$ (Theorem 1 of the manuscript). Note also that $\Delta P = 1/8$ is the largest transported gain any within-cell-constant prediction can earn here, since a proper score is maximised in expectation at the true frequencies $(3/4, 1/4)$ — which is what A predicts.

*Model B: a case-specific improvement.*. Cases 1–3 are unchanged, but case 4 now moves towards label 2, so $h_4(1) = -5/8$ and $h_4(2) = +3/8$. Every case is scored at $+3/8$ on its own label, so $\Delta T = 3/8$. Under transport cases 1–3 are unaffected, but case 4 is scored against three label-1 cases at $-5/8$ apiece, so $\Delta P = \frac{1}{4}[3 \cdot \frac{1}{24} - \frac{5}{8}] = -\frac{1}{8}$ and $\Delta R = \frac{1}{2}$.

Two things follow. First, B's aggregate gain of $3/8$ *understates* its case-level improvement of $1/2$, and not because B fits the cell frequencies worse — it fits them better, moving the cell-mean prediction from 0.5 to 0.625 against a truth of 0.75, an improvement of exactly 0.09375. What offsets it is dispersion: B's forecasts vary inside the cell where the old model's did not, and by Supplementary Proposition S2 the transported channel charges that variation at the same 0.09375, so the two cancel and the in-sample transported gain is zero. The $-1/8$ the estimator reports is what remains once each case is scored against the *other* cases' labels only, removing the $\frac{1}{n_c}\widehat{\Delta R} = 1/8$ the in-sample plug-in had credited to the transported channel. Second, an evaluator reading only the total would credit B with three times A's improvement, whereas the two improve on disjoint grounds: A earns $1/8$ purely by moving to the cell frequencies and gains no covariance, while B's $1/2$ of covariance is partly concealed by a transported term that turns negative. This is the failure mode that motivates the paper, in miniature — aggregate scores are not merely imprecise about composition, they can order two models by the wrong criterion.

*The covariance form, and why leaving one out matters..* Theorem 1 of the manuscript says $\Delta R$ is a within-cell covariance between the probability change and the label indicator. Computing it directly for model B gives $2\sum_y \mathrm{Cov}(\Delta q_y, \mathbb{1}\{Y = y\}) = 3/8$, not the $1/2$ transport returned. Neither is wrong: the covariance uses the cell's *own* four labels to form its reference, so each case contributes to the reference it is compared against, and that in-sample plug-in is attenuated by exactly $(n_c - 1)/n_c = 3/4$ — indeed $\frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$. Transport, which scores each case only against the *other* cases' labels, removes the attenuation with no held-out data. Proposition 1 of the manuscript shows this holds for every cell with $n_c \geq 2$ and every grouping variable, which is why the method needs no sample splitting. The example is reproduced as a unit test in the accompanying software.

One property the example cannot show, because both candidates happen to be equally confident, is the estimator's sensitivity to confidence itself. Multiply every prediction's odds by a constant and $\Delta R$ moves, though no case has been reordered and no information added; Section 4.1 of the manuscript measures that at $-0.488$ in a design where the two models carry identical case-level information. Every result in this paper is therefore reported for confidence-matched pairs.

*A decision the split changes..* At the benchmark's own label frequencies both candidates improve and B improves three times as much, so an evaluator reading

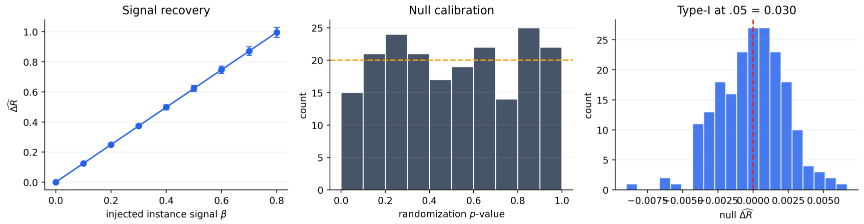


**Figure S1: Controlled validation.** WAGER recovers injected within-cell signal monotonically (mean ± one standard deviation), produces approximately uniform null randomization $p$-values, and controls type-I error in 200 null runs.

the score picks B, rightly. Suppose deployment sees label frequencies in this cell of $(\frac{1}{4}, \frac{3}{4})$ instead of the observed $(\frac{3}{4}, \frac{1}{4})$ — a label shift, the situation post-hoc prior correction addresses [1, 2]. Reweighting each case by $p'(y_i)/p(y_i)$ and rescoring sends A from $+\frac{1}{8}$ to $-\frac{3}{8}$ while leaving B at $+\frac{3}{8}$: A becomes *worse* than the model it was meant to replace, on a change that leaves every prediction exactly as it was. The decomposition says in advance which is exposed and the aggregate score does not, because A's whole gain sits in the transported channel — the one a change in label frequencies acts on — and B's in the covariance channel, which such a change leaves alone. That is the operational content of the split: $\widehat{\Delta P}$ is the part of a reported improvement contingent on the evaluation set's label frequencies matching deployment, and $\widehat{\Delta R}$ is the part that is not. Section 4.4 of the manuscript exhibits the same structure on real models.

## S2. Bregman-score identity

**Theorem S1** (Bregman-score covariance identity)**.** *Let* $H_x(y) = S_\psi(q_1(\cdot \mid x), y) - S_\psi(q_0(\cdot \mid x), y)$ *for a Bregman score generated by a strictly convex, differentiable* $\psi$*, and write* $\Delta g_y(x) = \partial_y\psi(q_1(\cdot \mid x)) - \partial_y\psi(q_0(\cdot \mid x))$ *for the contrast of the* $y$*-th gradient coordinate. Then*

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}\big(\Delta g_y(X),\ \mathbb{1}\{Y = y\} \mid C = c\big). \tag{S2}$$

**Corollary S1** (Quadratic and log score)**.** *For* $\psi(q) = \|q\|_2^2$*,* $\Delta g_y(x) = 2\Delta q_y(x)$ *and Eq.* (S2) *is Eq. (7) of the manuscript. For* $\psi(q) = \sum_k q_k \log q_k$*,* $\Delta g_y(x) = \log q_1(y \mid$

$x) - \log q_0(y \mid x)$, *and*

$$\Delta R_c = \sum_{y=1}^{K} \mathrm{Cov}\left(\log \frac{q_1(y \mid X)}{q_0(y \mid X)},\ \mathbb{1}\{Y = y\} \,\middle|\, C = c\right). \tag{S3}$$

Eq. (S3) gives the log-score check of Section 4.2 of the manuscript the same exact covariance meaning that Eq. (7) of the manuscript gives the quadratic score: within-group covariance under the log score is the within-cell covariance between the log-likelihood-ratio contrast of the two models and the observed label indicator, not merely "a sensitivity analysis." Any other Bregman score – the spherical or power-family scores catalogued by Gneiting and Raftery [3], for instance – inherits an identity of the same form through its own $\psi$, so Theorem S1 fixes the scope of the estimator's scoring-rule dependence once, rather than leaving each alternative score to be checked separately. The implemented log score floors each probability at $\epsilon = 10^{-6}$ to avoid $-\infty$ on an unseen label; Eq. (S3) is exact for the unfloored $\log q_y$, and the floored version used throughout is a sensitivity check on that exact identity rather than a literal instance of it, negligibly different whenever no relevant probability is within $\epsilon$ of zero.

## S3. Finite-sample identity

**Theorem S2** (Finite-sample WAGER identity)**.** *For every cell with $n_c \geq 2$, without a distributional assumption,*

$$\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c. \tag{S4}$$

*Moreover, when the cell's examples are drawn i.i.d. from the cell population, $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic for $\Delta R_c$. Independence, not exchangeability alone, is what unbiasedness requires: the independent-coupling definition of $\Delta P_c$ pairs an example with a label drawn independently from the same cell, so $\mathbb{E}[H_1(Y_2)]$ matches it only for independent within-cell draws.*

## S4. Randomization test

As a complementary finite-group diagnostic, WAGER applies an independently sampled cyclic shift to labels inside every cell. Each shift preserves cell size and label histogram. Recomputing $\widehat{\Delta R}$ under $B$ shifts gives

$$p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{\widehat{\Delta R}^{(b)} \geq \widehat{\Delta R}\}}{B+1}. \tag{S5}$$

This is exact under the sharp null that labels are exchangeable relative to the frozen model outputs within each cell. Because SGG relations share images, our primary real-data uncertainty statement is the image-cluster-robust interval; the randomization test is reported as a secondary relation-level diagnostic.

## S5. Coarsening and unrecorded covariates

Supplementary Section S10.4 reports that replacing the class-pair cell with a coarser subject-only cell changes the estimated covariance gain. The following result gives that sensitivity an exact population-level mechanism rather than treating it as an empirical curiosity.

**Proposition S1** (Coarsening decomposition)**.** *Let $\bar{\phi} = g(\phi)$ be any coarsening of the grouping variable, so that each cell $\bar{c}$ of $\bar{\phi}$ is a union of cells of $\phi$. Then*

$$\Delta R_{\bar{c}} = \mathbb{E}[\Delta R_C \mid \bar{\phi} = \bar{c}] + \sum_{y=1}^{K} \mathrm{Cov}\big(\mathbb{E}[H_X(y) \mid C],\, \Pr(Y = y \mid C) \,\big|\, \bar{\phi} = \bar{c}\big). \tag{S6}$$

The first term is the coarse cell's share of the fine-grained covariance gain; the second is a between-cell covariance, over the finer cells nested inside $\bar{c}$, between each fine cell's typical gain at label $y$ and its label-$y$ frequency. It is not sign definite, so coarsening $\phi$ can inflate or deflate the credited covariance gain, and it vanishes exactly when either quantity is constant within each coarse cell. This is why the finest defensible $\phi$ is the conservative default: coarsening adds cross-cell prior structure into $\Delta R$, while the finer partition's gain is preserved on average through the first term. The proof is a per-label application of the law of total covariance (Supplementary Section S1).

### *S5.1. Sensitivity to an unrecorded confounder*

Read in one direction, Proposition S1 says what folding a covariate into $\phi$ does. Read in the other, it quantifies what happens when a relevant covariate is *never* declared, which is the estimator's main scope limit. Let $Z$ be unrecorded and let $(\phi, Z)$ be the finer, fully adjusted cell. Applying Proposition S1 with $\bar{\phi} = \phi$ gives, for every declared cell $c$,

$$\Delta R_c(\phi) = \mathbb{E}\Big[\Delta R_{(\phi,Z)} \mid \phi = c\Big] + \sum_{y=1}^{K} \mathrm{Cov}(\mathbb{E}[H_X(y) \mid \phi, Z],\ \Pr(Y = y \mid \phi, Z) \mid \phi = c)\,, \tag{S7}$$

whose first term is what an analyst who could observe $Z$ would report and whose second is the bias of using $\phi$ alone. The identity is exact but needs $Z$'s joint law with $(H, Y)$. The following bound replaces that with a judgement about how strongly an unrecorded variable could plausibly act, in the spirit of Rosenbaum [4].

Supplementary Proposition S1 in Supplementary Section S5 states the bound: for a single elicited correlation $\bar{\rho}$ between an unrecorded variable's effect on the within-cell gain and its effect on the within-cell label distribution, the bias of using $\phi$ alone is at most $\bar{\rho}$ times a product of two dispersion factors, with a data-free form $\bar{\rho}M\sqrt{K-1}$ needing only the label cardinality and the score bound. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds ratio, it can be elicited the same way across benchmarks. Supplementary Section S5 evaluates both forms on our own estimates, and the sharp one is the less comfortable: an unrecorded covariate correlated at 0.061 would account for the entire covariance gain reported in Supplementary Section S10.2.

## S6. Proofs

### *S6.1. Proof of Theorem 1 of the manuscript*

Condition throughout on $C = c$. Expanding over labels gives

$$\Delta T_c = \sum_y \mathbb{E}[H_X(y)\mathbb{1}\{Y = y\}], \qquad \Delta P_c = \sum_y \mathbb{E}[H_X(y)] \Pr(Y = y).$$

Subtracting yields the covariance sum in Eq. (6) of the manuscript; the first identity is the definition of $\Delta R_c$. Under the quadratic score,

$$H_X(y) = 2\Delta q_y(X) - \{\|q_1(\cdot \mid X)\|_2^2 - \|q_0(\cdot \mid X)\|_2^2\}.$$

The bracketed term is independent of the label index. Its contribution summed over $y$ is $\mathrm{Cov}(B(X), \sum_y \mathbb{1}\{Y = y\}) = \mathrm{Cov}(B(X), 1) = 0$. The remaining terms give Eq. (7) of the manuscript. If $H_X(y) = H_c(y)$ almost surely, every covariance is zero. □

### *S6.2. Proof of Proposition S3*

Condition on $C = c$ and let $Y' \sim p(c)$ be independent of $X$. For the quadratic score $S_\mathrm{B}(q, y) = 2q_y - \|q\|^2$,

$$\mathbb{E}[S_\mathrm{B}(q_m, Y') \mid c] = 2\langle p(c), \bar{q}_m(c)\rangle - \mathbb{E}[\|q_m\|^2 \mid c]. \tag{S8}$$

Decompose the second moment as $\mathbb{E}[\|q_m\|^2 \mid c] = \|\bar{q}_m(c)\|^2 + \mathrm{tr}\,\mathrm{Var}(q_m \mid c)$, and complete the square using $2\langle p, \bar{q}\rangle - \|\bar{q}\|^2 = \|p\|^2 - \|p - \bar{q}\|^2$:

$$\mathbb{E}[S_\mathrm{B}(q_m, Y') \mid c] = \|p(c)\|^2 - \|p(c) - \bar{q}_m(c)\|^2 - \mathrm{tr}\,\mathrm{Var}(q_m \mid c). \tag{S9}$$

Subtracting the $m = 0$ expression from the $m = 1$ expression, the $\|p(c)\|^2$ terms cancel and Eq. (S17) follows, since $\Delta P_c = \mathbb{E}[S_\mathrm{B}(q_1, Y') \mid c] - \mathbb{E}[S_\mathrm{B}(q_0, Y') \mid c]$ by definition. □

*S6.3. Proof of Theorem S1*

Condition on $C = c$. The proof of Theorem 1 of the manuscript above establishes $\Delta R_c = \sum_y \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C = c)$ for *any* score contrast $H$, not only the quadratic one; only the identification of $H_x(y)$ with a specific functional form is score-dependent. For the Bregman score of the Bregman score $S_\psi(q, y) = \psi(q) + \langle \nabla\psi(q), e_y - q\rangle$,

$$H_x(y) = \underbrace{[\psi(q_1(x)) - \psi(q_0(x))] - [\langle \nabla\psi(q_1(x)), q_1(x)\rangle - \langle \nabla\psi(q_0(x)), q_0(x)\rangle]}_{=:B(x)} + \Delta g_y(x),$$

writing $q_i(x)$ for $q_i(\cdot \mid x)$, where $B(x)$ collects every term not indexed by $y$ and $\Delta g_y(x) = \partial_y\psi(q_1(x)) - \partial_y\psi(q_0(x))$ is the $y$-indexed remainder, exactly as in the proof of Theorem 1 of the manuscript for the quadratic case. Substituting,

$$\sum_y \mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid c) = \mathrm{Cov}\Big(B(X), \sum_y \mathbb{1}\{Y = y\} \Big| c\Big) + \sum_y \mathrm{Cov}(\Delta g_y(X), \mathbb{1}\{Y = y\} \mid c).$$

The first term is $\mathrm{Cov}(B(X), 1 \mid c) = 0$ since $\sum_y \mathbb{1}\{Y = y\} = 1$ almost surely. The second term is Eq. (S2). □

*S6.4. Proof of Theorem S2*

Summing the kernel in Eq. (11) of the manuscript over unordered pairs, every observed term $H_i(y_i)$ occurs in $n_c - 1$ pairs with coefficient $1/2$, while the two crossed terms together enumerate every ordered pair $H_i(y_j)$, $i \neq j$, with coefficient $1/2$. Since $\binom{n_c}{2} = n_c(n_c - 1)/2$,

$$\widehat{\Delta R}_c = \frac{1}{n_c}\sum_i H_i(y_i) - \frac{1}{n_c(n_c - 1)}\sum_{i \neq j} H_i(y_j) = \widehat{\Delta T}_c - \widehat{\Delta P}_c.$$

This is deterministic. When the cell's examples are i.i.d. draws from the cell population, the average of the symmetric order-two kernel is the standard unbiased U-statistic for its population expectation [5]; independence is used, not merely exchangeability, because $\Delta P_c$ is defined through an independent label coupling, so $\mathbb{E}[H_1(Y_2)] = \mathbb{E}[H_X(Y')]$ requires the two draws to be independent. $\square$

*S6.5. Proof of Proposition S1*

Fix $y$ and condition throughout on $\bar{\phi} = \bar{c}$. The law of total covariance applied to the pair $(H_X(y), \mathbb{1}\{Y = y\})$ with the finer conditioning variable $C$ nested inside $\bar{c}$ gives

$$\begin{aligned}\mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid \bar{\phi} = \bar{c}) &= \mathbb{E}[\mathrm{Cov}(H_X(y), \mathbb{1}\{Y = y\} \mid C) \bigm| \bar{\phi} = \bar{c}] \\ &\quad + \mathrm{Cov}(\mathbb{E}[H_X(y) \mid C],\ \Pr(Y = y \mid C) \bigm| \bar{\phi} = \bar{c}),\end{aligned}$$

using $\mathbb{E}[\mathbb{1}\{Y = y\} \mid C] = \Pr(Y = y \mid C)$. Summing over $y = 1, \ldots, K$ and applying the population identity of Theorem 1 of the manuscript to the left side (at $\bar{\phi} = \bar{c}$) and to the first term on the right side (at $C = c$, then averaged over $c$ within $\bar{c}$) yields Eq. (S6). $\square$

*S6.6. Proof of Proposition 1 of the manuscript*

Fix a cell $c$ with $n_c \geq 2$ and $i \in c$. By definition of the label count, $\sum_{y=1}^{K} m_{cy} H_i(y) = \sum_{j \in c} H_i(y_j) = H_i(y_i) + \sum_{j \neq i} H_i(y_j)$, and Eq. (12) of the manuscript gives $\sum_{j \neq i} H_i(y_j) = (n_c - 1)\hat{P}_i$. Hence

$$\tilde{P}_i = \frac{1}{n_c} \sum_{y} m_{cy} H_i(y) = \frac{H_i(y_i) + (n_c - 1)\hat{P}_i}{n_c}.$$

Substituting $H_i(y_i) = \hat{P}_i + \hat{R}_i$ gives $\tilde{P}_i = \hat{P}_i + n_c^{-1}\hat{R}_i$, and $\tilde{R}_i = H_i(y_i) - \tilde{P}_i = \hat{R}_i - n_c^{-1}\hat{R}_i = (n_c - 1)n_c^{-1}\hat{R}_i$. Averaging $\tilde{R}_i$ within cell $c$ and reweighting by $n_c$ across cells, as in the dataset-level definition of $\widehat{\Delta Z}$, gives $\widetilde{\Delta R} = N_*^{-1} \sum_c n_c \cdot \frac{n_c - 1}{n_c} \widehat{\Delta R}_c = N_*^{-1} \sum_c (n_c - 1)\widehat{\Delta R}_c$. $\square$

*S6.7. Proof of Proposition S2*

Fix $\phi = c$ throughout and write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$, both random through their dependence on $(\phi, Z)$. Let $U = (a_y - \mathbb{E}[a_y \mid c])_{y=1}^{K}$ and

$V = (b_y - \mathbb{E}[b_y \mid c])_{y=1}^K$ be the corresponding mean-zero $\mathbb{R}^K$-valued random vectors conditional on $\phi = c$. The map $(U, V) \mapsto \mathbb{E}[\langle U, V\rangle \mid c] = \sum_y \mathrm{Cov}(a_y, b_y \mid c)$ is an inner product on the space of mean-square-integrable $\mathbb{R}^K$-valued random variables conditional on $\phi = c$, so by the Cauchy–Schwarz inequality applied to that inner product,

$$\Big|\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)\Big| \le \sqrt{\mathbb{E}[\|U\|^2 \mid c]}\,\sqrt{\mathbb{E}[\|V\|^2 \mid c]} = \sqrt{\textstyle\sum_y \mathrm{Var}(a_y \mid c)}\,\sqrt{\textstyle\sum_y \mathrm{Var}(b_y \mid c)}.$$

By the definition of $\rho$, the left side equals $|\rho|\sqrt{\sum_y \mathrm{Var}(a_y \mid c)}\sqrt{\sum_y \mathrm{Var}(b_y \mid c)}$ exactly, so this step alone is not yet a bound; two further steps supply one, at two different levels of conservatism. First, $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$ is itself a conditional expectation of $H_X(y)$ given more information than $\phi = c$ alone, so the law of total variance gives $\mathrm{Var}(H_X(y) \mid c) = \mathbb{E}[\mathrm{Var}(H_X(y) \mid \phi, Z) \mid c] + \mathrm{Var}(a_y \mid c) \ge \mathrm{Var}(a_y \mid c)$; likewise $b_y = \Pr(Y = y \mid \phi, Z)$ satisfies $\mathrm{Var}(b_y \mid c) \le \mathrm{Var}(\mathbb{1}\{Y = y\} \mid c) = p_y(c)(1 - p_y(c))$ by the same argument applied to $\mathbb{1}\{Y = y\}$ in place of $H_X(y)$. Both bounds hold termwise, so summing over $y$ and applying the (monotone) square root to each sum separately gives the observable, per-cell bound

$$\sqrt{\textstyle\sum_y \mathrm{Var}(a_y \mid c)}\,\sqrt{\textstyle\sum_y \mathrm{Var}(b_y \mid c)} \le \sqrt{\textstyle\sum_y \mathrm{Var}(H_X(y) \mid c)}\,\sqrt{\textstyle\sum_y p_y(c)(1 - p_y(c))},$$

the second inequality of Eq. (S13); this is a bound on a product of two sums, and does not simplify to a sum of per-label square roots, since Cauchy–Schwarz runs the other way for that rearrangement. Second, for Corollary S2, $a_y \in [-M, M]$ as a conditional expectation of a $[-M, M]$-valued variable, so Popoviciu's inequality gives $\mathrm{Var}(a_y \mid c) \le M^2$ for every $y$ and hence $\sum_y \mathrm{Var}(a_y \mid c) \le KM^2$. For the $b_y$ the same argument would give $\mathrm{Var}(b_y \mid c) \le 1/4$ and a total of $K/4$, but that treats the $b_y$ as $K$ unconstrained quantities in $[0, 1]$ when in fact $\sum_y b_y = 1$ almost surely. Using that constraint, $\sum_y b_y^2 \le \left(\sum_y b_y\right)^2 = 1$ pointwise, while $\sum_y (\mathbb{E} b_y)^2 \ge$

$K^{-1}(\sum_y \mathbb{E}b_y)^2 = 1/K$ by Cauchy–Schwarz, so

$$\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c) = \mathbb{E}\Big[\sum_y b_y^2\Big] - \sum_y (\mathbb{E}b_y)^2 \le 1 - \frac{1}{K},$$

which is Eq. (S14) and is strictly smaller than $K/4$ for every $K > 2$. Therefore

$$\Big|\sum_{y=1}^{K} \mathrm{Cov}(a_y, b_y \mid c)\Big| \le |\rho|\,\sqrt{KM^2}\,\sqrt{1 - 1/K} = |\rho|\,M\,\sqrt{K-1}.$$

By Eq. (S6), the left side of each display is exactly $|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]|$, giving Eq. (S13) and Eq. (S15). □

## S7. Influence function: estimand and derivation

*The estimand under singleton exclusion..* Cells with $n_c = 1$ identify no within-cell counterfactual and are excluded, so the dataset-level statistic targets the *identified-subpopulation* quantity: writing $\pi_c$ for the cell probabilities, the target of $\widehat{\Delta R}$ is the $\pi$-weighted mean $\theta = \sum_c \pi_c\, \Delta R_c / \sum_c \pi_c$ taken over cells that appear with at least two examples. At finite $N$ the identified set is random; we treat the interval as conditional on identification and report the identified fraction alongside every estimate (99.0% of relations on VG150, 100% on CIFAR-100-LT), so the gap between the conditional and full-population targets is bounded by the reported coverage.

*Derivation of Eq. (14) of the manuscript..* Fix a cell $c$ and let $\theta_c = \Delta R_c$ with $g_c(z) = \mathbb{E}[A(z, Z') \mid Z' \in c]$ the kernel projection. The classical Hájek projection of the order-two U-statistic $\widehat{\Delta R}_c$ is

$$\widehat{\Delta R}_c = \theta_c + \frac{2}{n_c}\sum_{i \in c}(g_c(Z_i) - \theta_c) + O_p(n_c^{-1}).$$

The dataset statistic weights cells by their example counts, $\widehat{\Delta R} = N_*^{-1} \sum_c n_c \widehat{\Delta R}_c$, and cell membership is itself random (multinomial over cells), so an example $i$ contributes both through the within-cell projection and through the between-cell composition term $\theta_{C_i} - \theta$. Combining the two sources,

$$\widehat{\Delta R} - \theta = \frac{1}{N_*} \sum_i \Big[ 2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta) \Big] + o_p(N_*^{-1/2}),$$

where the dataset-level remainder requires one condition beyond the per-cell expansion, because $C$ per-cell remainders do not aggregate for free. Writing $C = \#\{c : n_c \geq 2\}$ for the number of eligible cells, the weighted sum of the per-cell degenerate terms is $N_*^{-1} \sum_c n_c \cdot O_p(n_c^{-1}) = O_p(C/N_*)$, which is $o_p(N_*^{-1/2})$ provided

$$C = o(\sqrt{N_*}). \tag{S10}$$

This is what makes the second-order rate $O_p(n_c^{-1})$ rather than $o_p(n_c^{-1/2})$ worth stating: the sharper per-cell rate is what allows a condition on the cell *count* alone, with no further requirement on individual cell sizes. On VG150, $C = 6{,}346$ against $N_* = 227{,}337$, so $C/\sqrt{N_*} \approx 13.3$ — the condition is a statement about the limit, and at these finite sample sizes it is the coverage simulation of Section 4.1 of the manuscript, not Eq. (S10), that supports the reported intervals. Estimating $g_{C_i}(Z_i)$ by the pair-kernel mean $\bar{A}_i$, $\theta_c$ by $\widehat{\Delta R}_c$, and $\theta$ by $\widehat{\Delta R}$ gives the plug-in influence contribution $\widehat{\psi}_i = 2(\bar{A}_i - \widehat{\Delta R}_c) + (\widehat{\Delta R}_c - \widehat{\Delta R}) = 2\bar{A}_i - \widehat{\Delta R}_c - \widehat{\Delta R}$, which is Eq. (14) of the manuscript and has exactly zero sample mean by construction. Its empirical 95% coverage in a VG-like regime (image-clustered examples, heavy-tailed cell sizes with a large mass of near-minimum cells) is validated in Section 4.1 of the manuscript.

*Proof of Theorem S3.*. The linearization above gives $\widehat{\Delta R} - \theta = N_*^{-1} \sum_i \psi_i + o_p(N_*^{-1/2})$, under Eq. (S10), for the population influence contribution $\psi_i = 2(g_{C_i}(Z_i) - \theta_{C_i}) + (\theta_{C_i} - \theta)$, which has zero mean by construction and satisfies $|\psi_i| \leq C(M)$ for an absolute constant $C(M)$ depending only on the score bound $M$ in condition (i), since $g_{C_i}$, $\theta_{C_i}$, and $\theta$ are themselves averages of the bounded kernel $A_{ij} \in [-2M, 2M]$. Because clusters are mutually independent (condition (ii)), $T_g = \sum_{i \in g} \psi_i$ for $g = 1, \ldots, G$ are independent, mean-zero random variables with $|T_g| \leq |g|\,C(M)$. Condition (ii)'s finite third moment for cluster sizes and $\max_g |g| = o(\sqrt{G})$ give $\sum_g E|T_g|^3 \leq C(M)^3 \sum_g \mathbb{E}|g|^3 = O(G)$ while $(\sum_g \mathrm{Var}(T_g))^{3/2} = \Theta(G^{3/2})$ under condition (iv), so Lyapunov's ratio $\sum_g E|T_g|^3/(\sum_g \mathrm{Var}(T_g))^{3/2} \to 0$; the Lyapunov condition implies the Lindeberg condition, and the Lindeberg–Feller central limit theorem for triangular arrays of independent summands [6, 7] gives $(\sum_g T_g)/\sqrt{\sum_g \mathrm{Var}(T_g)} \xrightarrow{d} \mathcal{N}(0,1)$. Condition (iii) makes $N_*/\sum_g \mathrm{Var}(T_g) \to 1/\sigma^2$ in probability by definition of $\sigma^2$, and the $o_p(N_*^{-1/2})$ remainder is asymptotically negligible after multiplying by $\sqrt{N_*}$, so $\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0,\sigma^2)$ by Slutsky's theorem. For the variance estimator, $\widehat{\Delta R} \to_p \theta$ by the weak law of large numbers across the diverging number of identified examples, and $\widehat{\Delta R}_c \to_p \Delta R_c$ for each eligible cell by the weak law of large numbers applied within that cell – this is where condition (iv) is used, and it is not implied by anything aggregate: (iv) guarantees $n_c \to \infty$ almost surely for every cell in the fixed, finite support of $\phi$, which is what a within-cell WLLN requires, whereas a condition on the total identified count bounds no individual cell's size. So $\widehat{\psi}_i \to_p \psi_i$ pointwise, and $\widehat{\sigma}^2$ – a sum of squares of cluster-aggregated $\widehat{\psi}_i$, normalized by $N_*$ and $G$ – is a continuous functional of these consistent plug-ins, hence $\widehat{\sigma}^2 \to_p \sigma^2$ by the continuous mapping theorem. □

*Efficient computation.*. The implementation does not instantiate all pairs. In addition to Eq. (12) of the manuscript, let $G_c(y) = \sum_{j\in c} H_j(y)$ and $O_c = \sum_{j\in c} H_j(y_j)$. The pair-kernel mean for observation $i$ is

$$\bar{A}_i = \frac{1}{2}\left[H_i(y_i) + \frac{O_c - H_i(y_i)}{n_c - 1} - P_i - \frac{G_c(y_i) - H_i(y_i)}{n_c - 1}\right].$$

Therefore both the point estimate and Hájek projection require only label counts, column sums, and matrix–vector products inside each cell. For image clusters $g$, the implemented variance is

$$\widehat{\mathrm{Var}}(\widehat{\Delta R}) = \frac{G}{G-1}\frac{1}{N_*^2}\sum_{g=1}^{G}\left(\sum_{i\in g}\widehat{\psi}_i\right)^2,$$

**Algorithm 1** WAGER gain decomposition

**Require:** frozen $q_1, q_0$; labels $y$; cells $\phi$; proper score $S$
**Ensure:** $\widehat{\Delta T}, \widehat{\Delta P}, \widehat{\Delta R}$ and uncertainty

$H_i(k) \leftarrow S(q_{1i}, k) - S(q_{0i}, k)$ for all $i, k$
**for** each cell $c$ with $n_c \geq 2$ **do**
  count labels $m_{c1}, \ldots, m_{cK}$
  **for** each $i \in c$ **do**
    $P_i \leftarrow (\sum_k m_{ck} H_i(k) - H_i(y_i))/(n_c - 1)$
    $R_i \leftarrow H_i(y_i) - P_i$
  **end for**
**end for**
$\widehat{\Delta T} \leftarrow \text{mean}\, H_i(y_i)$; $\widehat{\Delta P} \leftarrow \text{mean}\, P_i$; $\widehat{\Delta R} \leftarrow \text{mean}\, R_i$
compute image-clustered Hájek interval and cell-shift randomization $p$

where $G$ is the number of images and $\widehat{\psi}_i$ is Eq. (14) of the manuscript. We use a normal critical value; with 27,955 image clusters, small-$G$ corrections are immaterial.

### S8. Randomization algorithm

For each cell, sort examples once and index them $0, \ldots, n_c - 1$. Randomization $b$ draws $o_c^{(b)}$ uniformly from this index set and maps label position $r$ to $(r + o_c^{(b)}) \bmod n_c$. Independent offsets across cells define an element of the product of cyclic groups. The cell label histogram is unchanged, so $\sum_y m_{cy} H_i(y)$ in Eq. (12) of the manuscript is precomputed; only the assigned label lookup changes. One randomization is $O(N)$ after the $O(NK)$ setup. The plus-one Monte Carlo correction includes the observed assignment and prevents zero $p$-values.

### S9. Implementation and reproducibility

The NumPy implementation is in `wager/antisymmetric.py`. The primary routine `decompose_gain` validates probability matrices, forms score contrasts, excludes singleton cells, computes all three components, verifies the identity numerically, and returns per-instance transported/covariance values plus intervals. The Visual Genome driver is `experiments/run_sgg_wager.py`; controlled validation and sensitivity checks are in `experiments/antisymmetric_simulation.py` and `experiments/antisymmetric_ablations.py`. All random seeds are fixed. Unit tests cover the decomposition identity, quadratic covariance identity, cell-constant null, singleton reporting, cluster inference, randomization power, and log-score path, plus direct numerical checks of Propositions 1 of the manuscript and S1 (the latter against an explicit discrete population with exact expectations, independent of the estimator's own code path), Corollary S1's log-score covariance identity, Corollary 1 of the manuscript's equivalence to the (unclustered) Diebold–Mariano statistic at a trivial grouping variable, and Proposition S2's bound ordering (exact bias ≤ tight, per-cell bound ≤ crude, data-free bound) on a second explicit discrete population with a known omitted confounder. The covariance-identity cross-check quoted in Section S15.5 is reproduced by `experiments/cifar_covariance_check.py`; the calibration-matched protocol and the prior-matching experiment are `experiments/cifar_recalibration_control.py` and `experiments/vg_prior_consequence.py`. Table S2's free-bound rows are reproduced by `experiments/sensitivity_bound_check.py` and its sharp rows, including the robustness value $\rho^{\dagger}$ of Eq. (S11), by `experiments/sensitivity_analysis.py`; the evaluation of Corollary S2 is `experiments/sensitivity_bound_check.py`, which recomputes the crude bound directly from the committed `antisymmetric_ablations.json` rather than rerunning the decomposition; `experiments/sensitivity_analysis.py` computes Proposition S2's tighter, per-cell form from per-instance arrays (Section S9.1).

*Full-data VG predictors (Sections S15.1–S15.4)..* FREQ uses add-one smoothing over the 50 predicates within each training class pair, backing off to the subject-marginal and then the global predicate histogram for pairs unseen in training. The MLPs consume fixed random class embeddings (32 dimensions per class, seed 0, concatenated for subject and object) on standardized features and train with AdamW (learning rate $10^{-3}$, weight decay $10^{-5}$, batch size 4096): MLP-CLASS with hidden sizes (256, 128) for 12 epochs, MLP-SPATIAL with (256, 128) for 15 epochs, and MLP-SPATIAL+ with (512, 256, 128) for 20 epochs – the "+" variant differs in both width and schedule, as noted in Section 4.3 of the manuscript.

*Data availability..* The repository ships the estimator, every driver script, and the cached results files that every printed number traces to (`results/*.json`). The cached per-relation probability arrays that the drivers consume (`sgg_dists.npz`, `vg_visual_models.npz`, `cifar_lt_results.npz`) are regenerable from the committed training scripts with fixed seeds and are archived alongside the code release, so exact reproduction does not require retraining.

### *S9.1. Compute environment for Sections S15.3–S15.5*

Model training and feature extraction for both new experiments ran on a single NVIDIA T4 (15 GB) GPU. The WAGER decomposition itself is pure NumPy and requires no accelerator: it consumes the resulting cached probability arrays, as in Section 4.3 of the manuscript. All Python code runs under a pinned environment (Python 3.13).

*Matched-subsample models (Section 4.4 of the manuscript)..* The three compared predictors share a fixed, seeded (seed = 0) subsample of 100,000 of the 532,987 training relations, drawn without replacement uniformly at the relation level (not the image level); the full 229,605-relation test split is always used for evaluation. All three use the identical head and optimizer – AdamW, learning rate $10^{-3}$, weight decay $10^{-5}$, 15 epochs, batch size 4096, hidden sizes (256, 128), on standardized features – so they differ only in their input features.

Subject/object crops are taken from the annotated boxes and CLIP-preprocessed (bicubic resize to $224 \times 224$ with CLIP's published normalization constants); a crop with no valid overlap with the image falls back to a blank crop rather than being dropped, so every relation contributes a feature row. CLIP runs in half precision with a batch size of 512 crops and is never fine-tuned. Two implementation details are worth recording. First, the pretrained OpenAI weights were trained with the QuickGELU activation, so the encoder must be instantiated as `ViT-B-32-quickgelu`; loading the same weights under the plain `ViT-B-32` configuration silently substitutes a standard GELU and degrades every embedding. Second, because an annotated object box participates in several relations, crops are deduplicated by (image, box) before encoding and gathered afterwards: the 659,210 crops implied by the audited relations collapse to 422,143 unique ones, removing 36.0% of the encoder work without changing any feature. Only the images referenced by those relations are retrieved, individually rather than through the two bulk archives; of the 71,990 required, three were unavailable upstream and fall back to a blank crop.

*ResNet-32 triple (Section S15.5)..* CIFAR-100-LT uses the standard exponential-profile construction of Cui et al. [8] at imbalance ratio 100: class $c$ (zero-indexed, $0, \ldots, 99$) keeps $n_c = \lfloor 500 \cdot \mu^c \rfloor$ training images, where $\mu = 100^{-1/99}$, so class 0 keeps all 500 images and class 99 keeps 5; the balanced 10,000-image test split is untouched. All three models are a from-scratch, randomly initialized CIFAR ResNet-32 (three stages of five basic residual blocks, 16/32/64 channels) trained for 80 epochs with SGD (momentum 0.9, Nesterov, weight decay $2 \times 10^{-4}$), batch size 128, standard crop-and-flip augmentation, and a cosine learning-rate schedule (peak 0.1, five-epoch linear warmup). The only difference between the runs is the per-example loss weight. CE uses uniform weights throughout. CB applies the effective-number class-balanced weight $w_c \propto (1-\beta)/(1-\beta^{n_c})$ with $\beta = 0.9999$, renormalized to average to one across classes, from initialization. DRW trains with uniform weights until epoch 60 and switches to exactly the CB weights for the final twenty epochs [9]; no other hyperparameter changes at the switch. The robustness matrix of Table S9 repeats this recipe verbatim at seeds 1 and 2 (ratio 100) and at ratios 10 and 50 (seed 0), re-drawing the long-tailed subset with the configuration's own seed and initializing all three arms of a configuration identically (`experiments/colab_cifar_multiseed.py`). The post-hoc LA arm is not trained: it rescales the baseline's test probabilities by the inverse training class prior, $q_{\mathrm{LA}}(y \mid x) \propto q_{\mathrm{CE}}(y \mid x)/\pi_{\mathrm{train}}(y)$, renormalized.

## S10. Sensitivity bounds on the reported estimates

*What the bound says about this paper's own estimates..* Table S2 reports both forms of the bound on the Visual Genome pair of Section S15.2, together with the coarsening from the class-pair to the subject-only cell, which instantiates Eq. (S6) exactly with $Z$ = object class. The free bound is directionally correct and, at $K = 50$, still 1,887 times the actual coarsening bias, so any $\bar{\rho}$ an analyst would plausibly declare certifies a limit far above what is observed. That looseness is a property of a worst-case inequality applied to label probabilities that concentrate away from the extremes at which it binds, not a defect of this dataset — but it also means the free bound cannot support a claim about whether any particular reported $\widehat{\Delta R}$ would survive confounding.

The sharp per-cell form of Eq. (S13), which replaces the worst-case sums by the cell's own $\sum_y \mathrm{Var}(H_X(y) \mid c)$ and $\sum_y p_y(c)(1 - p_y(c))$, can support such a claim, and it is less reassuring than the free bound. Computed by `experiments/sensitivity_analysis.py` from per-instance arrays for the same pair, it gives $B = 0.23284$ at $\bar{\rho} = 1$, so the *robustness value* — the smallest aggregate correlation at which the bound reaches the estimate — is

$$\rho^{\dagger} = \frac{|\widehat{\Delta R}|}{B} = \frac{0.01427}{0.23284} = 0.06127. \tag{S11}$$

An unrecorded covariate whose aggregate correlation with the within-cell gain and the within-cell label distribution reaches 0.061 is enough to account for the whole of that $\widehat{\Delta R}$. At the $\bar{\rho} = 0.1$ that this section suggests as a plausible elicited value, the sharp bound is 0.02328, which *exceeds* the estimate. We therefore do not claim that a confounder would have to be implausibly strong to explain these Visual Genome gains away; on this evidence a modest one would suffice, and the reader should treat Section S15.2's covariance gains as conditional on the declared $\phi$ being close to complete. The flagship audit of Section 4.2 of the manuscript is not exposed in the same way, because its reported $\widehat{\Delta R}$ is indistinguishable from zero and a sensitivity bound on a null result changes nothing about it.

**Table S2:** Both forms of the Proposition S2 bound on the MLP-SPATIAL vs. MLP-CLASS pair, and the coarsening already reported in Table S7 ($K = 50$, $M = 2$ for the quadratic score). The empirical bias is the law-of-total-expectation aggregate of Eq. (S6)'s covariance term across all subject-only cells. `experiments/sensitivity_bound_check.py` reproduces the free-bound rows from the committed ablation results and `experiments/sensitivity_analysis.py` the sharp rows from per-instance arrays.

| Quantity | Value |
|---|---|
| $\widehat{\Delta R}$, $\phi$ = subject-only | 0.02181 |
| $\widehat{\Delta R}$, $\phi$ = class-pair | 0.01439 |
| Empirical coarsening bias | 0.00742 |
| Free bound, $M\sqrt{K-1}$, at $\bar{\rho} = 1$ | 14.00 |
| Minimum $\bar{\rho}$ for the free bound to reach the bias | 0.00053 |
| Sharp bound $B$ at $\bar{\rho} = 1$ | 0.23284 |
| Sharp bound at $\bar{\rho} = 0.1$ | 0.02328 |
| Robustness value $\rho^{\dagger}$ | 0.06127 |

## S11. Conditions for the limit theorem

Two things about condition (iv) deserve saying plainly rather than in a footnote. It is what licenses treating the per-cell plug-ins $\widehat{\Delta R_c}$ as consistent for $\Delta R_c$ in the variance argument, and a cell's size is not controlled by the aggregate identified count, so this needs its own divergence requirement. But it also forces every cell to be eventually identified, hence $N_*/N \to 1$: an earlier version of this theorem additionally assumed an identified fraction converging to some $\kappa \in (0, 1]$, which condition (iv) makes vacuous, and under which the identified-subpopulation estimand $\theta$ would in any case coincide with the unconditional target. We have therefore dropped that assumption rather than state two conditions that cannot both bind. The price is that the theorem does not describe the regime the paper's own applications sit in, where $\theta$ is a random, $N$-dependent target because coverage is short of one (99.0% on VG150, 98.9% in Section 4.2 of the manuscript). What the theorem licenses is the interval's form and its variance estimator; what licenses reading the reported intervals at finite $N$ is the coverage simulation of Section 4.1 of the manuscript, and Section 5 of the manuscript states the gap between the two rather than eliding it.

The proof (Section S6) combines the exact linearization of Section S7 with a Lindeberg–Feller central limit theorem for the independent-across-cluster sum of mean-zero, uniformly bounded terms [6, 7]; boundedness (i) makes the Lindeberg condition automatic given (ii), and consistency of $\widehat{\sigma}^2$ follows from the weak law of large numbers applied to each plug-in $\widehat{\Delta R}_c, \widehat{\Delta R}$ together with continuity of the variance functional. One step needs more care than that section's per-cell statement supplies: a per-cell Hájek remainder of order $o_p(n_c^{-1/2})$ does not aggregate to $o_p(N_*^{-1/2})$ at the dataset level when the number of cells grows, since summing $C$ such remainders admits a $\sqrt{C}$ inflation. The repair does not need a new condition on cell sizes, only that the cell count be negligible relative to the identified sample, $\#\{c : n_c \geq 2\} = o(N_*)$: the second-order degenerate term of a two-sample U-statistic is $O_p(n_c^{-1})$ rather than $o_p(n_c^{-1/2})$, and $\sum_c n_c \cdot O_p(n_c^{-1}) = O_p(C) = o_p(N_*)$ gives the dataset-level rate directly. We state that condition explicitly; on VG150 it holds comfortably ($C = 6{,}346$ against $N_* = 227{,}337$). Condition (ii) is what rules out a single cluster dominating the sum – an image contributing a positive fraction of all relations, for instance – and is a standard requirement for cluster-robust inference generally, not specific to this estimator. This theorem licenses reading the image-clustered intervals reported throughout Section 4 of the manuscript as asymptotically valid in the idealized regime where every eligible cell's size grows without bound. That regime is not attained in-sample: VG150's eligible cells are heavy-tailed, and 2,286 of the 6,346 eligible class-pair cells hold fewer than five examples (Table S7). The proportion is smaller than earlier drafts of this paper asserted — those cells carry 2.7% of identified relations, and the median eligible cell holds appreciably more than the minimum — but a third of the cells still sit where condition (iv)'s divergence is plainly hypothetical. The reported 94.0% empirical coverage against a nominal 95% (Section 4.1 of the manuscript), not this theorem directly, is therefore the evidence that the interval remains a good approximation well short of the idealization.

### S12. Choice of the grouping variable

*Fineness is not a total order, and the guidance needs that qualification..* The advice to prefer the finest defensible $\phi$ is sound against *coarsening*, which is what

Proposition S1 governs. It gives no guidance at all between two partitions of the same granularity, neither of which coarsens the other — and there the estimand can differ without limit. Four cases suffice. Take two labels, $q_0$ uninformative at $(\frac{1}{2}, \frac{1}{2})$, and a new model that is confidently right on all four: $q_1 = (0.9, 0.1)$ on the two cases labelled 1 and $(0.1, 0.9)$ on the two labelled 2. The total gain is $\widehat{\Delta T} = +0.480$ under every partition, and every partition below is fully identified at 100% coverage:

| declared $\phi$ | cells | $\widehat{\Delta P}$ | $\widehat{\Delta R}$ |
|---|---|---|---|
| $\{1,3\},\{2,4\}$ (labels differ within each cell) | $2 \times 2$ | $-1.120$ | $+1.600$ |
| $\{1,2,3,4\}$ (one cell) | $1 \times 4$ | $-0.587$ | $+1.067$ |
| $\{1,2\},\{3,4\}$ (labels agree within each cell) | $2 \times 2$ | $+0.480$ | $0.000$ |

The first and third partitions are equally fine, both fully identified, and they disagree about whether this model pair's improvement is entirely case-level or entirely transported. The third is the degenerate case the paper already warns about — grouping by something that determines the label leaves each cell label-homogeneous and drives $\widehat{\Delta R}$ to zero algebraically — but the warning was previously attached to the choice $\phi = Y$, and the point here is more general and less comfortable: any $\phi$ correlated with the label inside a cell partially degenerates the estimand, "finest defensible" cannot detect it, and Proposition S1 is silent because neither partition is a coarsening of the other. What the analyst actually needs to declare, and what we recommend reporting alongside $\phi$, is the within-cell label heterogeneity that remains: a partition whose cells are nearly label-homogeneous cannot identify much of anything, and the coverage statistic does not reveal it. This is a limitation of the construction rather than of any application of it, and we do not have a procedure that resolves it.

*A recalibration-invariant alternative, and what it would cost..* Because $\Delta R$ is a covariance between forecast values, its sensitivity to recalibration is structural rather than incidental, and a reader may reasonably ask why we do not use a rank-based within-cell contrast instead — a within-cell paired AUC difference, say, for which DeLong et al. [10] already supply the correlated-U-statistic variance. Such a quantity would be invariant to any monotone rescaling of either model and would dissolve the confound this paper spends a protocol on. We think the trade is worth stating and is not obviously in our favour on every axis. What a rank-based contrast gives up is the exact identity: it is not a component of the score difference, so it cannot be added to anything to recover $\Delta T$, and a benchmark reporting it learns how the models rank cases but not how the reported score gain decomposes. The construction here buys exact additivity with respect to a declared proper score, and pays for it with scale dependence. A rank-based diagnostic is the natural companion rather than a competitor, and reporting both — one exact and scale-dependent, one invariant and non-additive — is what we would now recommend to a benchmark that could choose.

**Proposition S2** (Sensitivity bound for an unrecorded confounder)**.** *Suppose $H_x(y) \in [-M, M]$ for all $x, y$ (WAGER already requires a bounded score). Fix a cell $c$ of $\phi$, write $a_y = \mathbb{E}[H_X(y) \mid \phi, Z]$, $b_y = \Pr(Y = y \mid \phi, Z)$ (both conditional on $\phi = c$) for the two $K$-vectors on the right of Eq.* (S6)*, and let*

$$\rho = \frac{\sum_{y=1}^{K} \operatorname{Cov}(a_y, b_y \mid c)}{\sqrt{\sum_{y=1}^{K} \operatorname{Var}(a_y \mid c)}\sqrt{\sum_{y=1}^{K} \operatorname{Var}(b_y \mid c)}} \in [-1, 1] \tag{S12}$$

*be their aggregate correlation. Then*

$$\begin{aligned}\left|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\right| &\le |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \operatorname{Var}(a_y \mid c)}\ \sqrt{\textstyle\sum_{y=1}^{K} \operatorname{Var}(b_y \mid c)} \\ &\le |\rho| \sqrt{\textstyle\sum_{y=1}^{K} \operatorname{Var}(H_X(y) \mid \phi = c)}\ \sqrt{\textstyle\sum_{y=1}^{K} p_y(c)(1 - p_y(c))},\end{aligned} \tag{S13}$$

*where $p_y(c) = \Pr(Y = y \mid \phi = c)$; the second inequality holds termwise (*$\operatorname{Var}(a_y \mid c) \le \operatorname{Var}(H_X(y) \mid c)$ *and* $\operatorname{Var}(b_y \mid c) \le p_y(c)(1 - p_y(c))$ *by the law of total variance, since conditioning on more variables cannot increase the variance of a conditional expectation) followed by monotonicity of the square root applied to each sum separately – a product of two square-root sums, not a sum of per-label* *square roots, which the law of total variance alone would not license.*

**Corollary S2** (Free bound)**.** *Each $a_y$ ranges over $[-M, M]$ as a conditional expectation of $H$, so $\mathrm{Var}(a_y \mid c) \le M^2$ and $\sum_y \mathrm{Var}(a_y \mid c) \le KM^2$. The label probabilities satisfy $\sum_y b_y = 1$, which gives $\sum_y b_y^2 \le 1$ and, by Cauchy–Schwarz, $\sum_y (\mathbb{E} b_y)^2 \ge 1/K$; hence*

$$\sum_{y=1}^{K} \mathrm{Var}(b_y \mid c) = \mathbb{E}\Big[\textstyle\sum_y b_y^2\Big] - \textstyle\sum_y (\mathbb{E} b_y)^2 \;\le\; 1 - \tfrac{1}{K}. \tag{S14}$$

*Substituting both into Eq.* (S13) *gives the data-free bound*

$$\Big|\Delta R_c(\phi) - \mathbb{E}[\Delta R_{(\phi,Z)} \mid \phi = c]\Big| \;\le\; |\rho|\, M \sqrt{K-1}. \tag{S15}$$

Earlier versions of this corollary bounded $\sum_y \mathrm{Var}(b_y \mid c)$ by Popoviciu's inequality applied label by label, giving $K/4$ and hence $|\rho| KM/2$. That discards Eq. (S14): the $b_y$ are not $K$ unconstrained quantities in $[0, 1]$ but a point on the simplex, and $K/4$ is unattainable for $K > 2$. At VG150's $K = 50$ and $M = 2$ the corrected free bound is $14.00|\rho|$ rather than $50.00|\rho|$, a factor of 3.57.

An analyst willing to bound $|\rho| \le \bar{\rho}$ – how strongly an unrecorded variable could jointly move a model's within-cell score gain and the within-cell label distribution – obtains a numeric limit on how far $\widehat{\Delta R}_c$ can be from what a fully adjusted audit would find, without observing or modeling $Z$, in the spirit of Rosenbaum [4] and recent partial-correlation sensitivity models for unmeasured confounding [11]. Because $\bar{\rho}$ is a correlation rather than a domain-specific odds or risk ratio, it can be elicited the same way across benchmarks.

Both forms of the bound are evaluated on this paper's own Visual Genome estimates in Section S10. The short version is that the free bound is directionally correct but far too loose to certify anything, while the sharp per-cell form is *less* reassuring than the free one: an unrecorded covariate whose aggregate correlation with both channels reaches 0.061 would account for the entire covariance gain reported in Section S15.2. We therefore do not claim those gains are robust to confounding, and Section 4.2 of the manuscript's audit is unaffected only because its estimate is already indistinguishable from zero.

**Theorem S3** (Asymptotic normality of the covariance-gain estimator)**.** *Suppose (i) $H_x(y) \in [-M, M]$ for all $x, y$; (ii) examples are partitioned into clusters $g = 1, \ldots, G$ (e.g. images) that are mutually independent, with cluster sizes having a finite third moment and $\max_g |g| = o(\sqrt{G})$; (iii) the limiting variance $\sigma^2 = \lim_{G\to\infty} \mathrm{Var}(\sqrt{N_*}(\widehat{\Delta R} - \theta))$ exists and is strictly positive, where $\theta$ is the identified-subpopulation estimand of Section S7; and (iv) $\phi$ takes values in a fixed, finite set, each with strictly positive limiting population mass, so that every eligible cell's size $n_c$ diverges almost surely as $N \to \infty$. Then*

$$\sqrt{N_*}(\widehat{\Delta R} - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2), \tag{S16}$$

*and the plug-in sandwich variance $\widehat{\sigma}^2 = N_* \dfrac{G}{G-1} \sum_g \big(\sum_{i\in g} \widehat{\psi}_i / N_*\big)^2$ of Section S7 is consistent for $\sigma^2$.*

Section S11 discusses what condition (iv) costs — it forces every cell to be eventually identified, so the theorem does not describe the finite-sample regime our own applications sit in — and states the additional condition on the number of cells that the dataset-level remainder requires. What licenses reading the reported intervals at finite $N$ is the coverage simulation of Section 4.1 of the manuscript, not this theorem.

### S13. Terminology

### S14. Composition of the transported gain

The component that survives transport is easy to misname. It is tempting to call $\Delta P$ a prior-fit gain, since transport replaces a case's label by an independent draw from its cell; several earlier drafts of this paper did exactly that. The following identity shows the name is wrong, and says what the quantity is instead.

**Table S3:** The paper's vocabulary, and what it corresponds to in standard terms.

| Term used here | Standard counterpart | Meaning |
|---|---|---|
| grouping variable $\phi$ | stratifying / conditioning variable | a discrete function of the input, declared before the audit, whose levels the labels depend on |
| cell $c$ | level of $\phi$; stratum | the set of test cases sharing one value of $\phi$ |
| label transport | within-stratum relabelling; permutation within blocks | replacing a case's label with another case's label from the same cell |
| total gain $\Delta T$ | paired score differential | the observed mean score of the new model minus that of the old |
| transported gain $\Delta P$ | no exact counterpart; a mean-forecast-fit term minus a dispersion term (Proposition S3), *not* a measure of prior fit | the part of $\Delta T$ that survives label transport |
| within-group covariance gain $\Delta R$ | the covariance term of a covariance rearrangement of a proper score [12]; twice a classical resolution difference *only* under within-cell calibration | the part destroyed by transport; a within-cell prediction–label covariance |
| coverage | identified fraction | share of test cases in cells of size $\geq 2$, the only cells in which $\Delta R$ is identified |
| calibration-matched | temperature-scaled on held-out data | both models rescaled to comparable confidence before the audit |

**Proposition S3** (Composition of the transported gain)**.** *Under the quadratic score, write* $\bar{q}_m(c) = \mathbb{E}[q_m(\cdot \mid X) \mid C = c]$ *for model m's mean forecast in cell c,* $\mathrm{Var}(q_m \mid c)$ *for the within-cell covariance matrix of its forecast vector, and* $p(c)$ *for the cell's transport law. Then*

$$\Delta P_c = \underbrace{\Big[\|p(c) - \bar{q}_0(c)\|^2 - \|p(c) - \bar{q}_1(c)\|^2\Big]}_{\textit{improvement in mean-forecast fit}} - \underbrace{\Big[\mathrm{tr}\,\mathrm{Var}(q_1 \mid c) - \mathrm{tr}\,\mathrm{Var}(q_0 \mid c)\Big]}_{\textit{increase in within-cell dispersion}}. \tag{S17}$$

Only the first bracket is prior fit: it is positive exactly when the new model's average within-cell forecast sits closer to the cell's label frequencies. The second has nothing to do with the prior. It rewards the new model for making *less* dispersed predictions inside the cell, at a fixed mean forecast, so $\Delta P$ can be large and positive when neither model's prior fit has changed at all. One cell suffices:

with two labels and frequencies $(\frac{1}{2}, \frac{1}{2})$, let $q_0$ alternate between $(1, 0)$ and $(0, 1)$ in a pattern uncorrelated with the label and let $q_1$ predict $(\frac{1}{2}, \frac{1}{2})$ throughout. Both mean forecasts equal the cell frequencies exactly, so the first bracket is zero, and the whole observed gain of $\frac{1}{2}$ is transported, produced entirely by the collapse in dispersion. The channel is therefore the reliability-and-sharpness channel of the classical partition, not a prior-fitting channel, which is why we name it for the operation. Two consequences: a claim that some gain is "mostly prior-fitting" does not follow from a large $\widehat{\Delta P}$ alone, and Eq. (S17) explains why confidence matching moves $\widehat{\Delta P}$ so sharply, since a temperature change is almost purely a change in dispersion.

## S15. Supplementary studies

This section reports the two studies summarized in Section 4.5 of the manuscript and the granularity, score and cell-size sensitivity analyses referred to there. Each applies the estimator of Section 3 of the manuscript without modification; only the declared grouping variable $\phi$ changes.

### *S15.1. Controlled predictors on VG150 in full*

We work with Visual Genome [13] in the PredCls regime [14], where ground-truth boxes and object labels are supplied and the model has only to predict one of 50 predicate labels for each ordered object pair. The predictors of this section are audited on a VG150-*style* dataset we build directly from the released relationship annotations: the 150 most frequent object classes and 50 most frequent predicates form the vocabulary, and images are split 70/30 into a pool for fitting the compared predictors and a pool on which the audit runs. We use this reconstruction rather than the canonical VG150 release because these predictors are trained here from annotations alone; Section 4.2 of the manuscript audits *released third-party checkpoints* on the canonical split itself, so both settings appear in the paper and neither is asked to stand for the other. The reconstruction provides 532,987 training and 229,605 test relations, drawn from 65,228 and 27,955 images respectively,

**Table S4:** Frozen-model top-1 accuracy on VG150 PredCls. Similar accuracies do not reveal whether the difference is transported.

| Model | Accuracy |
|---|---|
| FREQ | 0.6564 |
| FREQ+OVERLAP | 0.6545 |
| MLP-CLASS | 0.6554 |
| MLP-SPATIAL | 0.6639 |
| MLP-SPATIAL+ | 0.6674 |

and the cell throughout is $\phi$ = (subject class, object class), two real examples of which appear in Figure S2. Of the 8,614 cells present in the test split, 6,346 contain at least two relations and are therefore identified; the remaining 2,268 are singletons, which WAGER excludes and reports rather than smoothing over, leaving 227,337 relations (99.0%) under audit.

The five predictors are deliberately constructed so that each exposes a different kind of information. FREQ is the training-count distribution conditioned on the class pair, with add-one smoothing and a subject-marginal and then global backoff for class pairs unseen in training; FREQ+OVERLAP adds to it a single binary box-overlap indicator. MLP-CLASS consumes fixed subject and object class embeddings, which makes it a function of $\phi$ alone; MLP-SPATIAL extends those embeddings with 12 normalized box-geometry features covering relative centre, scale, aspect, intersection, containment and IoU, while MLP-SPATIAL+ both widens the network and trains longer (Section S9 lists every architecture and schedule), so its increment over MLP-SPATIAL reflects capacity and optimization jointly rather than capacity alone. Each is trained once on the training split, and WAGER thereafter consumes nothing but their cached test distributions, scored quadratically with image-clustered intervals and 499 within-cell cyclic randomizations, whose smallest attainable $p$-value is $1/500 = .002$; table entries of .002 are at this granularity floor and should be read as $p \leq .002$. With confidence intervals as the primary uncertainty statement and every significant effect many standard errors from zero, we do not additionally adjust the diagnostic $p$-values for the number of reported comparisons.

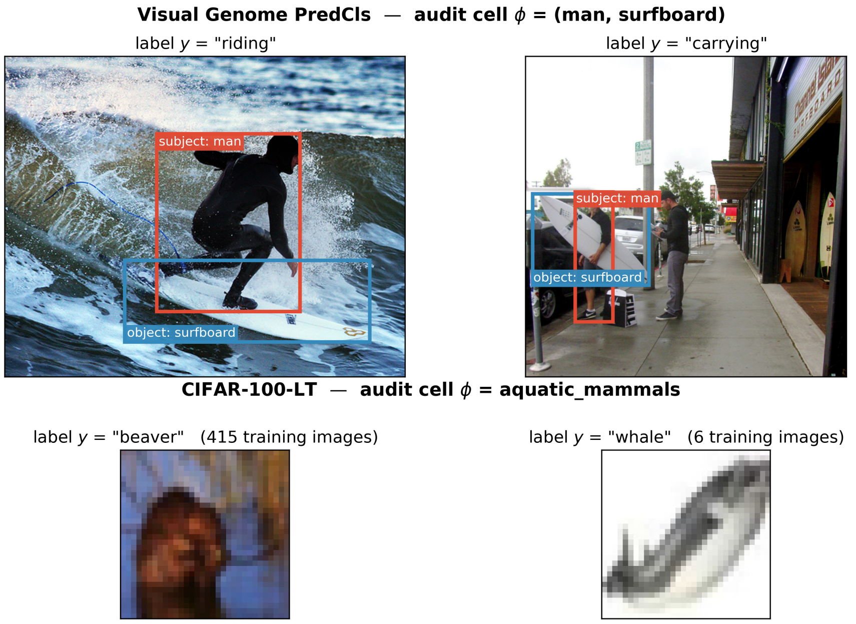


**Figure S2: What a cell contains, on both benchmarks.** Each row shows two real test examples that share the same declared grouping variable $\phi$ but carry different labels – precisely the configuration within-cell transport exploits. *Top:* two Visual Genome relations from the cell $\phi$ = (man, surfboard). The class pair is identical and therefore cannot distinguish "riding" from "carrying"; only the image content can, which is what $\Delta R$ is designed to credit. Boxes are the annotated subject (red) and object (blue) regions that MLP-SPATIAL turns into geometry features and MLP-VISUAL crops and encodes with CLIP. *Bottom:* two CIFAR-100 test images from the superclass $\phi$ = aquatic_mammals, drawn from opposite ends of the long-tailed training distribution (415 versus 6 training images). Crossing the labels within a row preserves the cell and its label histogram while destroying which prediction belongs with which instance.

### *S15.2. Gain attribution for the controlled predictors*

The central result is collected in Table S5 and Figure S3. MLP-CLASS improves on FREQ by 0.03386 of quadratic score, yet because its gain vector is constant inside every class-pair cell, the estimator assigns that entire improvement to prior transport and precisely nothing to instance covariance. The zero is worth dwelling on: it follows algebraically from the construction rather than from a learned baseline happening to pass a sanity check.

Adding box geometry changes the picture. MLP-SPATIAL gains 0.04270 over FREQ, of which 0.02832 survives label transport and 0.01439 is within-group covariance (CI $[0.01373, 0.01504]$). More revealing still is the direct comparison against MLP-CLASS, where the total gain falls to 0.00885 only because the transported channel deteriorates by 0.00554 even as the covariance gain from spatial features rises by 0.01439. The resulting covariance share of 1.63 records a genuinely useful instance-specific change that the aggregate score partly conceals behind a worse prior component, and it is exactly the configuration that a ratio confined to $[0, 1]$, or thresholded at 0.5, would misdescribe.

The remaining two comparisons bracket the interesting range. MLP-SPATIAL+ adds 0.00374 over MLP-SPATIAL, nearly all of it covariance gain at 0.00321 (CI $[0.00289, 0.00353]$), whereas FREQ+OVERLAP supplies the opposite caution: its overlap indicator does create a small positive covariance gain, but the accompanying negative prior term is larger, so the model ends up worse overall. Using image-derived information is thus never treated as an improvement in itself; both channels are reported, signs included.

*Reading the magnitudes..* Quadratic-score units are not ones most readers calibrate by eye, so it is worth anchoring them in the rank-based metrics this literature reports. Where both channels point the same way the correspondence is straightforward: MLP-SPATIAL-S's $\widehat{\Delta R} = +0.01023$ over MLP-CLASS-S accompanies +0.51 points of top-1 predicate accuracy, +0.60 of mean reciprocal rank and +0.86 of recall@5, so in these data a hundredth of quadratic-score gain is worth roughly half an accuracy point. The covariance channel is not, however, a re-description of any rank metric, and the real-pixel comparison of Section 4.4 of the manuscript is where that becomes visible: there $\widehat{\Delta R} = +0.00641$ while top-1 accuracy falls by 3.07 points and recall@5 by 2.21. Prior-matching both models before scoring narrows the accuracy gap to 2.17 points without closing it. The rank metrics aggregate the two channels into one number; when the channels carry opposite signs, as they do here, no single such number can represent both, which is precisely the situation the decomposition exists to report.

*Reading $\widehat{\Delta T}$ as a comparative forecast test..* Corollary 1 of the manuscript identifies $\widehat{\Delta T}$ and its image-clustered interval in Table S5 as a grouped Diebold–Mariano statistic and interval, on the identified subsample, for each pair's paired score differential, so every row already reports what that classical test would re-

**Table S5:** WAGER decomposition on 227,337 identified VG150 PredCls relations. Gains use the quadratic score. CI is image-cluster robust; $p$ is the 499-shift relation-level randomization diagnostic (granularity floor .002). "–" denotes an undefined share because total gain is negative. The MLP-CLASS row's zero-width interval is exact by construction, not a sampling statement: a gain vector constant within every cell makes the antisymmetric kernel identically zero (Theorem 1 of the manuscript), so no variability exists for the interval to measure.

| New model | Old model | $\widehat{\Delta T}$ | $\widehat{\Delta P}$ | $\widehat{\Delta R}$ (95% CI) | $\widehat{\rho}$ | $p$ |
|---|---|---|---|---|---|---|
| FREQ+OVERLAP | FREQ | −0.01229 | −0.01430 | 0.00202 [0.00168, 0.00235] | – | .002 |
| MLP-CLASS | FREQ | 0.03386 | 0.03386 | 0.00000 [0.00000, 0.00000] | 0.00 | .724 |
| MLP-SPATIAL | FREQ | 0.04270 | 0.02832 | 0.01439 [0.01373, 0.01504] | 0.34 | .002 |
| MLP-SPATIAL+ | FREQ | 0.04644 | 0.02885 | 0.01760 [0.01686, 0.01834] | 0.38 | .002 |
| MLP-SPATIAL | MLP-CLASS | 0.00885 | −0.00554 | 0.01439 [0.01373, 0.01504] | 1.63 | .002 |
| MLP-SPATIAL+ | MLP-SPATIAL | 0.00374 | 0.00053 | 0.00321 [0.00289, 0.00353] | 0.86 | .002 |

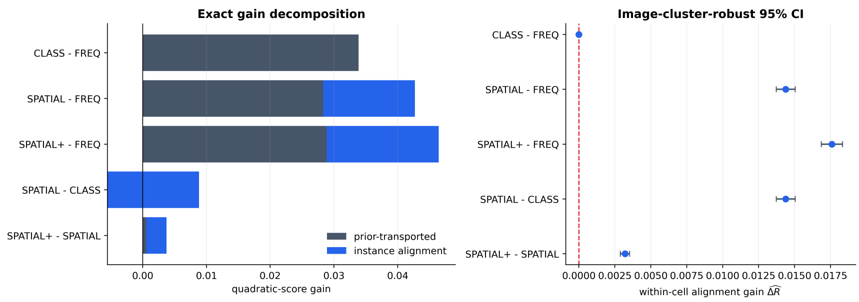


**Figure S3: Direct model-pair attribution.** Left: exact decomposition of each positive gain into transported and within-group covariance components (the negative prior term for SPATIAL–CLASS extends left of zero). Right: primary covariance estimates with image-cluster-robust intervals.

port. What it does not report is the next two columns: whether MOTIFS-TDE (Section 4.2 of the manuscript) or MLP-SPATIAL beat their baseline for a reason a frequency-only remodeling of the same benchmark could not reproduce, which is the question $\widehat{\Delta P}$ and $\widehat{\Delta R}$ answer and DM/GW do not ask.

### *S15.3. Real pixels against a geometric proxy, in full*

None of the predictors above ever sees an image: they are functions of the class pair and, for MLP-SPATIAL(+), of box coordinates recorded in the annotation. Since box geometry is only a proxy for image content, a sceptical reader may ask whether the covariance channel responds to pixel evidence or merely to that proxy. Figure S2 sharpens the question — the cell $\phi$ = (man, surfboard) holds both "riding" and "carrying", and what distinguishes them is what the photograph shows.

MLP-VISUAL answers it. Subject and object regions are cropped from the original image and each encoded by a frozen CLIP ViT-B/32 encoder [15]; the two 512-dimensional embeddings are concatenated with the class embeddings MLP-CLASS already uses and passed to a small trained head. Because CLIP is never fine-tuned, whatever covariance the model gains reflects information its pretraining already recovered from pixels rather than representation learning on Visual Genome.

Encoding every training relation with CLIP is expensive, so MLP-VISUAL must be trained on a subsample; setting a subsampled model against the full-data MLP-SPATIAL would confound feature content with training-set size. All three compared predictors are therefore retrained on one fixed, seeded subsample of 100,000 of the 532,987 training relations, written MLP-CLASS-S, MLP-SPATIAL-S and MLP-VISUAL-S, so any gap between them isolates the features. The audit still uses the full 229,605-relation test split with the same class-pair $\phi$ and image-clustered intervals.

*Result.*. By any aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6161 against 0.6467 for MLP-SPATIAL-S and 0.6416 for MLP-CLASS-S, and a quadratic score 0.04655 below MLP-SPATIAL-S. An evaluator reading either number would discard it.

The decomposition contradicts that reading on the axis that matters. Against MLP-SPATIAL-S the CLIP model loses 0.05296 of transported gain while *gaining* 0.00641 of covariance (95% CI $[0.00514, 0.00769]$, randomization $p = .002$): its entire deficit sits in the transported channel and its within-cell discrimination is significantly better, not worse. Against the class-only baseline the point is plainer still — pixels buy 0.01665 of covariance where box geometry buys 0.01023, with well-separated intervals. Real image content carries more case-specific signal than the geometric proxy standing in for it.

A plausible mechanism, stated as hypothesis rather than finding: the 1024 CLIP dimensions dominate the 64 dimensions of class embedding the head also receives, so the model fits VG150's class-pair frequency structure markedly less well than a predictor built around it. A benchmark whose headline metric rewards

**Table S6:** Matched-subsample real-pixel study on the full VG150 test split (coverage 99.0%, as in Table S5). All three predictors are trained on the same 100,000 relations and differ only in features. CI is image-cluster robust on $\widehat{\Delta R}$; $p$ is the 499-shift randomization diagnostic.

| New model | Old model | $\widehat{\Delta T}$ | $\widehat{\Delta P}$ | $\widehat{\Delta R}$ (95% CI) | $p$ |
|---|---|---|---|---|---|
| MLP-SPATIAL-S | MLP-CLASS-S | 0.00503 | −0.00520 | 0.01023 [0.00967, 0.01079] | .002 |
| MLP-VISUAL-S | MLP-CLASS-S | −0.04152 | −0.05816 | 0.01665 [0.01534, 0.01796] | .002 |
| MLP-VISUAL-S | MLP-SPATIAL-S | −0.04655 | −0.05296 | 0.00641 [0.00514, 0.00769] | .002 |

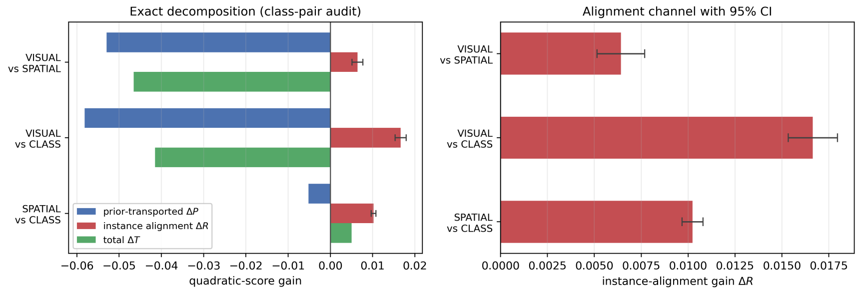


**Figure S4: Pixels beat geometry on covariance while losing on the aggregate.** Left: the exact decomposition for the three matched-subsample pairs. Both comparisons involving the CLIP model have strongly negative totals (green) driven entirely by the transported channel (blue), while their covariance channel (red) is positive in every case. Right: the covariance channel alone, with 95% image-clustered intervals. Replacing box geometry with real image content raises within-group covariance from 0.01023 to 0.01665 over the same class-only baseline.

transported gain will punish this model, and the aggregate numbers duly do. What the decomposition adds is that the punishment is not for failing to use the image. Two falsification checks, reported in Section S9, support that reading: correcting a model's within-cell prior towards the training histogram registers as almost pure transported movement (+0.02401 against +0.00111 of covariance), as a correction carrying no case-level information must, and applied to the CLIP model it halves the aggregate deficit (−0.04655 to −0.02143) while the covariance advantage persists (+0.00752, and +0.00663 when both models are corrected) and survives confidence matching (+0.00467, CI [+0.00314, +0.00621]).

### *S15.4. Sensitivity analyses*

For MLP-SPATIAL versus MLP-CLASS, the conclusion survives every planned sensitivity check (Table S7). Changing from the quadratic to a probability-floored log score changes scale (0.01439 to 0.06536) but not sign or significance. The coverage column matters for reading the rows against each other, and earlier versions of this table omitted it. Each row targets the identified

**Table S7:** Sensitivity of the MLP-SPATIAL vs. MLP-CLASS decomposition to score, audit granularity, and minimum cell size. CI is image-cluster robust on $\widehat{\Delta R}$.

| Factor | Setting | $\widehat{\Delta T}$ | $\widehat{\Delta R}$ (95% CI) | Cells | Cov. |
|---|---|---|---|---|---|
| Score | Brier (def.) | 0.00885 | 0.01439 [0.01373, 0.01504] | 6346 | 99.0% |
| Score | Log (floored) | 0.04297 | 0.06536 [0.06294, 0.06779] | 6346 | 99.0% |
| Cell $\phi$ | class pair (def.) | 0.00885 | 0.01439 [0.01373, 0.01504] | 6346 | 99.0% |
| Cell $\phi$ | subject only | 0.00943 | 0.02181 [0.02079, 0.02283] | 150 | 100.0% |
| Min. cell size | 2 (def.) | 0.00885 | 0.01439 [0.01373, 0.01504] | 6346 | 99.0% |
| Min. cell size | 5 | 0.00772 | 0.01395 [0.01329, 0.01462] | 4060 | 96.3% |
| Min. cell size | 10 | 0.00683 | 0.01330 [0.01262, 0.01398] | 2744 | 92.6% |
| Min. cell size | 20 | 0.00570 | 0.01225 [0.01155, 0.01295] | 1757 | 86.7% |

subpopulation its own setting defines, so the minimum-cell-size rows are not four estimates of one quantity: raising the minimum from 2 to 20 drops coverage from 99.0% to 86.7% and changes the estimand along with the estimator. That the estimate moves so little across them (0.01439 to 0.01225) is therefore mild evidence of stability rather than a sensitivity analysis holding the target fixed.

Coarsening $\phi$ from the class pair to subject class increases the credited covariance to 0.02181. Proposition S1 identifies why: the coarser cell's estimate equals the class-pair estimate's conditional average plus a between-class-pair covariance term, and that term is positive here, meaning some of the credited gain reflects class-pair-level structure that the finer audit had already correctly assigned to $\Delta P$. This is precisely the sense in which the class-pair audit is the conservative choice: it is not that coarsening is wrong, but that it cannot be assumed neutral, and only the finer partition's estimate isolates covariance gain net of every prior regularity expressible in $\phi$. The same class-pair-versus-subject-only pair also instantiates Proposition S2 directly, with the subject class playing the role of the unrecorded confounder $Z$: Table S2 in Section S10 reports that the resulting bias of 0.00742 is comfortably inside the Cauchy–Schwarz bound, though the bound is loose enough at $K = 50$ that it would remain satisfied for a $\bar{\rho}$ three orders of magnitude smaller than one. Raising the minimum cell size from 2 to 5, 10, and 20 yields 0.01395, 0.01330, and 0.01225; all cluster-robust lower limits remain positive. The procedure itself has no smoothing strength, split fraction, or decision threshold to tune.

### *S15.5. Long-tailed image classification*

Sections 4.3–4.4 of the manuscript are all instances of one task (predicate classification on an ordered object pair). Section 2.4 of the manuscript argued that long-tailed image classification poses a structurally identical question: a class-frequency prior can be fit better without any per-instance improvement. We test this directly on CIFAR-100-LT [8], an exponential-imbalance (ratio 100) subsample of CIFAR-100's training split with the original, balanced test split retained for evaluation. We train ResNet-32 [16] classifiers that differ only in their per-class loss weighting: a vanilla cross-entropy baseline (CE); class-balanced effective-number re-weighting applied from initialization (CB) [8]; and the same re-weighting deferred to the final twenty epochs (DRW) [9]. Optimizer, schedule, augmentation, epoch budget, and seed are identical across runs (Section S9).

*Choosing $\phi$.*. A declared grouping variable has to predict the label without determining it, and this second benchmark makes the constraint unusually easy to violate. Were $\phi$ taken to be the fine class label itself, every cell would be label-homogeneous, within-cell transport would have nothing to cross, and $\widehat{\Delta R}$ would come out identically zero – an algebraic artifact wearing the appearance of a finding. CIFAR-100's own 20 coarse superclasses avoid this, each gathering five fine labels into a cell of 500 balanced test images, which is structurally the situation of a VG class-pair cell holding several predicates (Figure S2). Alongside the superclass audit we report two further declared priors: the trivial coarsening to a single global cell, which audits against the global class marginal and so instantiates Proposition S1 in a second domain, and a partition into head, medium and tail training-frequency tiers. CIFAR-100 test images are independent, unlike the several relations VG draws from one photograph, so no cluster grouping is required and the plain Hájek interval of Section 3.9 of the manuscript applies directly. One semantic point matters in this domain: transport uses each cell's *test* label histogram, which on CIFAR-100-LT is balanced. A positive $\Delta P$ here therefore records movement *toward* the balanced test marginal – undoing the training-frequency skew – which is precisely the adjustment long-tailed evaluation is designed to reward; a transported gain is a description of where score was earned, not an accusation (Section 5 of the manuscript).

*Result.*. The three classifiers reach top-1 accuracies of 0.3662 (CE), 0.2627 (CB) and 0.3874 (DRW), so re-weighting from initialization degrades the classifier substantially while the deferred schedule delivers the improvement it was designed to produce. Table S10 decomposes both gains against the superclass audit.

Of the two, the class-balanced run is the more instructive, precisely because its aggregate score conceals almost everything of interest. On the quadratic score it gains +0.00143 over the baseline, a difference small enough that a reader working from the headline number alone would reasonably conclude that re-weighting had changed very little. The decomposition tells another story: within superclasses the transported channel improves by +0.19713 while within-group covariance deteriorates by −0.19569 (95% CI [−0.20728, −0.18411]), so the near-zero total is not a small effect but two large ones that happen to cancel.

*A calibration-matched regime.*. A channel split of that shape could, however, be produced by confidence scaling alone. Under the quadratic score the covariance channel is a covariance between probability movements and labels (Eq. (7) of the manuscript), and a monotone recalibration moves it: softening an overconfident model transfers score mass from the covariance channel to the transported channel while adding no information about any individual image. The baseline invites exactly this concern, assigning a mean maximum probability of 0.764 while being correct on only 0.3662 of the test set. We therefore repeat the audit in a calibration-matched regime: each model's temperature is chosen by held-out likelihood on a calibration half of the test split ($T^* = 2.85$ for CE, 3.51 for CB, 2.61 for DRW), the decomposition is computed on the disjoint audit half, and the whole procedure is repeated over twenty random splits (Table S8). The regime's necessity is quantified by its control row: recalibrating the baseline alone – a transformation carrying no instance information – moves the channels by +0.380 (prior) and −0.175 (covariance), so the raw split does mix calibration with dis-

**Table S8:** Raw and calibration-matched decompositions on CIFAR-100-LT at the superclass audit. Temperatures are fitted by likelihood on a held-out calibration half of the test split; every decomposition uses the disjoint audit half; entries are means (sd) over twenty random splits. The final row is the control: recalibrating the baseline alone carries no instance information yet moves both channels substantially.

| Comparison | Regime | $\widehat{\Delta T}$ | $\widehat{\Delta P}$ | $\widehat{\Delta R}$ |
|---|---|---|---|---|
| CB vs CE | raw | +0.00208 (0.00642) | +0.19635 (0.00369) | −0.19427 (0.00632) |
| CB vs CE | calibration-matched | −0.12912 (0.00373) | +0.11079 (0.00372) | −0.23991 (0.00491) |
| DRW vs CE | raw | +0.06712 (0.00865) | +0.04258 (0.00463) | +0.02454 (0.00685) |
| DRW vs CE | calibration-matched | +0.02173 (0.00275) | +0.00216 (0.00410) | +0.01957 (0.00329) |
| CE recalibrated vs CE | control | +0.20495 (0.00420) | +0.37960 (0.00268) | −0.17464 (0.00455) |

crimination.

The calibration-matched rows then separate the two schedules cleanly, and more sharply than the raw ones. CB's covariance deficit does not shrink but widens, to −0.23991 (0.00491): the model re-weighted from initialization has genuinely lost within-superclass discrimination – a real degradation that the ten-point accuracy drop records independently – and its near-zero raw total reflects that loss offset by its softer, better-calibrated confidence. DRW inverts the reading its raw split suggests. Raw, its +0.06712 gain divides roughly two-thirds transported to one-third covariance gain; calibration-matched, the improvement is +0.02173 (0.00275) of which +0.01957 (0.00329) is within-group covariance while the transported channel is close to zero (+0.00216 (0.00410)). What survives calibration matching is precisely the component the deferred schedule was designed to buy – per-instance discrimination, concentrated on rare classes (Table S11) – while the apparently transported share of its raw gain was largely the calibration difference between the two runs. Neither an accuracy difference nor an aggregate score exposes either composition, which is the practical argument for reporting the two channels, in both regimes, instead of their sum.

*Seeds, imbalance ratios, and a post-hoc arm..* A decomposition computed from one trained pair carries only test-set sampling uncertainty, which is not the uncertainty relevant to a claim about a training *method*. We therefore retrained the triple at two further seeds at the primary ratio and at imbalance ratios 10 and 50, and added a fourth arm: post-hoc logit adjustment (LA), which divides the baseline's predictions by the training class prior without retraining [17]. Every configuration is audited in both regimes; Table S9 reports the calibration-matched covariance channel, which is the one the preceding paragraphs interpret.

Three things follow. First, the class-balanced run's covariance loss is not a misconfiguration but a systematic function of imbalance severity: −0.069 at ratio 10, −0.244 at 50, and −0.251 (0.011) across three seeds at 100, tracking an accuracy penalty that grows from 2.3 to 10.4 points over the same range. Second, the deferred schedule's covariance gain is positive at every ratio and in every seed, but its magnitude varies roughly fourfold across seeds at ratio 100 (+0.007 to +0.031). We therefore report the spread rather than a seed-level interval: the direction replicates, the magnitude is seed-sensitive, and the gap between the tight within-run intervals of Table S10 and this across-seed spread is exactly why a method-level claim needs replication rather than a single audit.

Third, LA is an instructive falsification of a natural expectation. A purely post-hoc prior correction might be expected to register as almost pure transported channel, as the within-cell prior matching of Section 4.4 of the manuscript does. It does not: calibration-matched, LA reads as almost entirely covariance gain (+0.022 to +0.038) with a transported channel near zero, while achieving the best balanced accuracy of any arm at every ratio. The reason is visible in the construction. Dividing by the *global* class prior is not a within-superclass histogram move, because fine classes inside a superclass differ in training frequency, and the correction it applies is multiplicative in each example's own predicted probabilities — it changes rankings where the model already carried latent evidence. The two post-hoc corrections in this paper therefore land in opposite channels, and correctly so: transport separates adjustments that act on the cell's label histogram from adjustments that act on individual predictions.

Where the covariance gain sits is visible in Table S11 and Figure S5 (raw regime, primary seed). For DRW it falls where the long-tail literature intends it to, reaching +0.09009 on few-shot classes and +0.06760 on medium-shot ones, while many-shot classes lose −0.07624 as the schedule shifts capacity away from the head. The class-balanced run reproduces that same ordering with every level

**Table S9:** Calibration-matched covariance gain across seeds and imbalance ratios, with balanced test accuracy for each arm. LA is post-hoc logit adjustment applied to the baseline. The class-balanced deficit scales with imbalance severity; the deferred schedule's gain is positive throughout but seed-sensitive in magnitude; the post-hoc correction lands in the covariance channel rather than the transported channel.

| | | Top-1 accuracy | | | | $\widehat{\Delta R}$ vs. CE (calibration-matched) | | |
|---|---|---|---|---|---|---|---|---|
| Ratio | Seed | CE | CB | DRW | LA | CB | DRW | LA |
| 10 | 0 | 0.5388 | 0.5154 | 0.5447 | 0.5497 | −0.06903 | +0.00617 | +0.02222 |
| 50 | 0 | 0.4256 | 0.3242 | 0.4399 | 0.4550 | −0.24409 | +0.01903 | +0.03768 |
| 100 | 0 | 0.3662 | 0.2627 | 0.3874 | 0.3959 | −0.23936 | +0.01987 | +0.03449 |
| 100 | 1 | 0.3614 | 0.2287 | 0.3762 | 0.3877 | −0.26176 | +0.00748 | +0.02672 |
| 100 | 2 | 0.3687 | 0.2481 | 0.3968 | 0.4024 | −0.25213 | +0.03124 | +0.03601 |
| 100, mean (sd) | | | | | | −0.25108 (0.01124) | +0.01953 (0.01188) | +0.03241 (0.00498) |

**Table S10:** WAGER decomposition on the 10,000-image balanced CIFAR-100-LT test split. Gains use the quadratic score; CI is the Hájek interval on $\widehat{\Delta R}$; $p$ is the 499-shift randomization diagnostic, one-sided for a positive covariance gain, so the value of 1.000 on the CB rows records a covariance gain at the extreme negative end of its randomization distribution.

| Comparison | Audit cell $\phi$ | $\widehat{\Delta T}$ | $\widehat{\Delta P}$ | $\widehat{\Delta R}$ (95% CI) | $p$ |
|---|---|---|---|---|---|
| CB vs CE | superclass (20) | 0.00143 | 0.19713 | −0.19569 [−0.20728, −0.18411] | 1.000 |
| CB vs CE | global (1) | 0.00143 | 0.25573 | −0.25430 [−0.26801, −0.24058] | 1.000 |
| CB vs CE | tier (3) | 0.00143 | 0.25247 | −0.25103 [−0.26453, −0.23754] | 1.000 |
| DRW vs CE | superclass (20) | 0.06606 | 0.04206 | 0.02400 [0.01340, 0.03460] | .002 |
| DRW vs CE | global (1) | 0.06606 | 0.03522 | 0.03085 [0.01860, 0.04309] | .002 |
| DRW vs CE | tier (3) | 0.06606 | 0.03702 | 0.02904 [0.01702, 0.04107] | .002 |

displaced downward, its many-shot covariance collapsing to −0.41977. What the decomposition supplies, then, is not only how much of a long-tail gain is real but where in the frequency spectrum it was earned.

These numbers also provide an implementation check on real inputs. Computing the within-group covariance gain directly from the identity of Eq. (7) of the manuscript, through a code path that shares nothing with the transport estimator, gives −0.19530; multiplying by the factor $n_c/(n_c - 1)$ that Proposition 1 of the manuscript predicts, at the $n_c = 500$ of a balanced superclass cell, returns −0.195693 against the estimator's −0.19569. The identities of Theorem 1 of the manuscript and Proposition 1 of the manuscript are algebraic, so this verifies the implementation rather than the theory; the point of the check is that two independent code paths agree to the fifth decimal on real trained models, not only on the synthetic populations used in the unit tests.

Coarsening behaves as Proposition S1 predicts. Moving from the 20 superclass cells to a single global cell changes the credited covariance from −0.19569

**Table S11:** Covariance gain by training-frequency tier at the primary superclass audit. The deferred schedule concentrates its genuine per-instance improvement on rare classes; both methods lose covariance on the head.

| | | CB vs CE | DRW vs CE | |
|---|---|---|---|---|
| Tier | $n$ | $\widehat{\Delta R}$ | $\widehat{\Delta T}$ | $\widehat{\Delta R}$ |
| Few-shot (< 20) | 3000 | −0.03337 | 0.13785 | 0.09009 |
| Medium-shot (20–100) | 3500 | −0.11075 | 0.10799 | 0.06760 |
| Many-shot (> 100) | 3500 | −0.41977 | −0.03740 | −0.07624 |

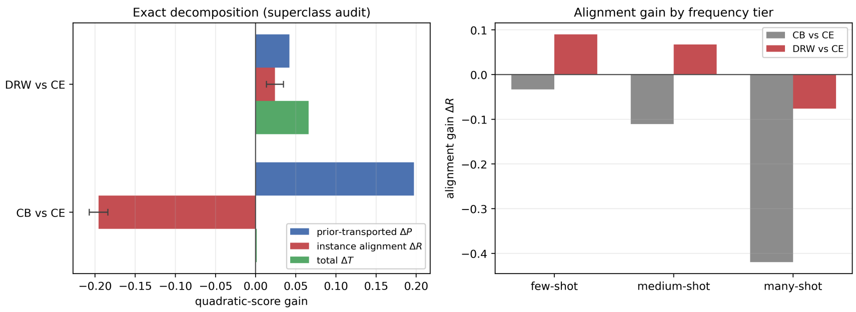


**Figure S5: The same attribution transfers to a second task.** Left: exact decomposition at the superclass audit. The CB total gain (green) is visually almost absent at +0.00143 while its two channels reach ±0.2 – a near-zero aggregate score concealing two large opposing effects. DRW's smaller total is genuinely split between both channels. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: the covariance channel by training-frequency tier, showing that DRW's genuine per-instance gain is concentrated on rare classes and negative on the head.

to −0.25430; the difference is the between-cell covariance term, which is negative here, so the coarse audit overstates the covariance loss by attributing to it structure that the superclass partition correctly assigns to $\Delta P$. The finer partition is again the conservative reading, for the same reason as in Section S15.4 and now in a second domain.

### *S15.6. Long-tailed text classification*

Sections S15.2–S15.5 test two vision tasks that share cached probability vectors and grouping variables grounded in class frequency or object geometry. To test the breadth Section 5 of the manuscript claims – any benchmark supplying two models' full probability vectors, labels, and a declared discrete grouping variable – we apply the identical estimator, unmodified, to document category prediction on 20 Newsgroups [18]. Neither the predictors nor the grouping variable are visual here: documents replace images, and the declared $\phi$ is a topical grouping rather than a spatial or class-frequency one, so a favorable result cannot be attributed to anything specific to vision data.

We build a long-tailed training subset with the same exponential profile as CIFAR-100-LT (imbalance ratio 100, Section S15.5), applied to 20 Newsgroups' 20 fine categories, keeping the standard balanced test split (7,532 documents) untouched (Section S9). The 20 categories fall into six conventional coarse topics – computing, recreation, science, politics, religion, and classifieds – which we use as the primary declared $\phi$, the direct analogue of CIFAR's coarse superclasses; a global (trivial) and a training-frequency-tier partition are reported alongside, as in Section S15.5. Documents are represented by a TF-IDF vector reduced to 300 dimensions by truncated SVD, and the three compared classifiers – CE, CB, and DRW – follow the same three-arm protocol as Section S15.5: an identical small MLP architecture and optimizer across runs, differing only in per-example loss weight (Section S9). Test documents are independent units, so the plain Hájek interval of Section 3.9 of the manuscript applies directly, as for CIFAR.

*Result..* Table S12 decomposes both re-weighting schedules' gain over the cross-entropy baseline (top-1 accuracies 0.3532 CE, 0.3960 CB, 0.3776 DRW). The two schedules land in opposite channels here, which is itself the finding worth reporting: on this task it is class-balanced re-weighting from initialization, not the deferred schedule, that buys genuine within-topic covariance gain (+0.04317, CI $[0.03664, 0.04970]$, 37.9% of its total gain), while DRW's comparably sized total gain (0.11490 vs. CB's 0.11388) is 98.6% transported, its residual covariance statistically indistinguishable from zero at the superclass audit (+0.00155, CI $[-0.00296, 0.00605]$, $p = .236$) and significantly negative once the coarser tier partition is used instead ($-0.00901$, CI $[-0.01428, -0.00374]$). This is the reverse of Section S15.5's image-classification finding, where CB's near-zero total masked a large covariance-gain *loss* and DRW's gain was genuinely covariance-driven. Nothing about the estimator or the construction of $\phi$ changed between the two applications; what changed is which training recipe, on which data, buys real per-instance discrimination – exactly the composition question an aggregate

**Table S12:** WAGER decomposition on the 7,532-document balanced 20-Newsgroups-LT test split. Gains use the quadratic score; CI is the (unclustered) Hájek interval on $\widehat{\Delta R}$, documents being independent units; $p$ is the 499-shift randomization diagnostic, one-sided for a positive covariance gain.

| Comparison | Audit cell $\phi$ | $\widehat{\Delta T}$ | $\widehat{\Delta P}$ | $\widehat{\Delta R}$ (95% CI) | $p$ |
|---|---|---|---|---|---|
| CB vs CE | superclass (6) | 0.11388 | 0.07071 | 0.04317 [0.03664, 0.04970] | .002 |
| CB vs CE | global (1) | 0.11388 | 0.07203 | 0.04185 [0.03381, 0.04990] | .002 |
| CB vs CE | tier (3) | 0.11388 | 0.07810 | 0.03578 [0.02825, 0.04331] | .002 |
| DRW vs CE | superclass (6) | 0.11490 | 0.11335 | 0.00155 [−0.00296, 0.00605] | .236 |
| DRW vs CE | global (1) | 0.11490 | 0.11809 | −0.00319 [−0.00884, 0.00245] | .980 |
| DRW vs CE | tier (3) | 0.11490 | 0.12391 | −0.00901 [−0.01428, −0.00374] | 1.000 |

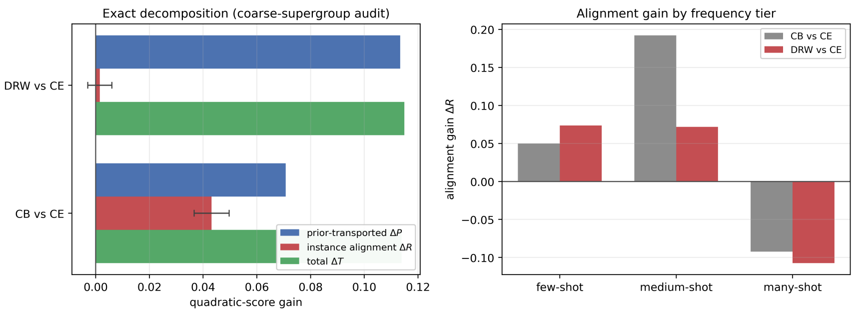


**Figure S6: A third domain, with neither a spatial nor a class-frequency prior.** Left: exact decomposition at the coarse-topic audit; CB's total gain is genuinely split between both channels while DRW's is almost entirely transported. Error bars are 95% Hájek intervals on $\widehat{\Delta R}$. Right: covariance gain by training-frequency tier for both schedules, showing the same head-loses/tail-gains signature as Figure S5 despite the change of domain and grouping variable.

score or an accuracy difference alone cannot answer, now demonstrated on a domain with neither a class-frequency table nor a spatial prior.

Coarsening moves both comparisons in the direction Proposition S1 allows without requiring: CB's credited covariance falls from 0.04317 at the superclass partition to 0.04185 at the global cell and 0.03578 at the tier partition, while DRW's changes sign twice, from a small positive residual (superclass) to a small negative one (global) to a significant negative one (tier). None of these differences overturns CB's conclusion, but DRW's illustrates exactly why the finest defensible partition is the conservative default (Section 3.5 of the manuscript): a reader stopping at the coarser tier partition would report a significant covariance *loss* for a method whose finer-grained residual is not distinguishable from zero.

Table S13 breaks the superclass-audit covariance down by training-frequency tier for both schedules; the two methods agree in shape even though their super-

**Table S13:** Covariance gain by training-frequency tier at the superclass audit, 20-Newsgroups-LT. Both schedules gain within-group covariance on rare and medium categories and lose it on frequent ones.

| Tier | $n$ | $\widehat{\Delta R}$, CB vs CE | $\widehat{\Delta R}$, DRW vs CE |
|---|---|---|---|
| Few-shot (< 20) | 1954 | 0.04991 | 0.07352 |
| Medium-shot (20–100) | 2610 | 0.19213 | 0.07171 |
| Many-shot (> 100) | 2968 | −0.09227 | −0.10753 |

class totals diverge sharply in composition, each gaining covariance on rare and medium-frequency categories and losing it on frequent ones, the same qualitative signature Section S15.5 reports for image classification.

This third domain uses a single trained pair per schedule rather than the multi-seed, calibration-matched protocol Section S15.5 applies to CIFAR-100-LT; extending that fuller protocol to text is the natural next replication, not a step this section claims to have taken.

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